

Week 2 · Laboratory Solutions

- These are **reference** answers, not the only correct ones. The checker tests the response, not the diagram.
- Read them after attempting the problems.

Building and checking

```
cd lectures/W02_simulink/solutions
W02_S1_pid_loop          % builds W02_S1.slx – all three problems in one model
W02_S_expected           % regenerates the expected-result figures

W02_check(1,'W02_S1')
W02_check(2,'W02_S1')
W02_check(3,'W02_S1')
```

All three pass and `check_overlaps('W02_S1')` is **0**. The output below is the measured one.

```
W02 problem 1
steady value at Kp = 2          0.5000 (expected 0.5000 +- 0.005 m) PASS
overshoot at Kp = 2            16.3055 (expected 16.3000 +- 0.5 %) PASS
steady value at Kp = 10        0.8334 (expected 0.8330 +- 0.005 m) PASS
overshoot at Kp = 10           38.7670 (expected 38.8000 +- 0.5 %) PASS

W02 problem 2
built by hand, no PID or Derivative PASS (PID blocks 0, Derivative blocks 0)
overshoot                      0.1580 (expected 0.1600 +- 0.3 %) PASS
inside 1 % of the setpoint after 2.7710 (expected 2.7700 +- 0.1 s) PASS
largest force                   129.6013 (expected 129.6000 +- 1.5 N) PASS

W02 problem 3
largest force                   2.5000 (expected 2.5000 +- 1e-06 N) PASS
overshoot, Kb = 0              28.6731 (expected 28.6700 +- 0.5 %) PASS
overshoot, Kb = 2              0.0316 (expected 0.0300 +- 0.3 %) PASS
settling (2 %), Kb = 0         5.5430 (expected 5.5400 +- 0.1 s) PASS
settling (2 %), Kb = 2         3.5610 (expected 3.5600 +- 0.1 s) PASS
```

One model for three problems

Problem 1 is the finished law with $K_i = K_d = 0$; problem 2 is the finished law with $\tau_{\max}$ large; problem 3 is the finished law. `W02_check` assigns the gains by name, so one model built to the final equation passes all three. Building it once, in full, is the reason the gains must be names rather than numbers.

$$e = y_d - y, \quad u = K_p e + I + d, \quad d = N_f(K_d e - \dot{x}), \quad \dot{x} = d, \quad \dot{I} = K_i e + K_b(\tau - u), \quad \tau = \text{sat}(u)$$

The choices in the answer, and why

	The choice	Why the alternative is worse
The derivative is a loop, not a block	gain N_f with an integrator in its feedback path	a Derivative block turns the corner of the step into an impulse and every step of sensor noise into a spike (§2-7); the loop form has the ceiling $K_d N_f$ built in
The excess is $\tau - u$, taken after the limit	a Sum with τ on its plus input and u on its minus input	taken the other way round the sign is wrong and the "anti-windup" winds the integrator faster; taken before the limit it is always zero
Rows P, D, I from the top	the integrator's junction at the bottom	the back-calculation line then comes up into it from below without crossing any other line; <code>check_overlaps</code> is 0
The plant sits at the height of the controller's output	the force enters it on one straight segment	the lines to the plant and to the log would otherwise cross the plant block

Why problem 2's force is 129.6 N

At the instant of the step the error jumps from 0 to 1 m. The proportional branch jumps to $K_p \cdot 1 = 10$ N. The filtered derivative of a unit jump starts at $K_d N_f = 6 \cdot 20 = 120$ N and decays with time constant $1/N_f = 50$ ms. Together, 130 N.

The checker measures 129.6 N, and the 0.4 N is the solver, not the law. The fourth stage of the `ode4` step that ends at $t = 1$ s already sees the step, so by the first logged sample the filter state has moved by $(h/6) \cdot 120 = 0.02$, and $d = N_f(K_d - x) = 20(6 - 0.02) = 119.6$ N. The log confirms $d = 119.6000$ N at $t = 1.0000$ s.

Nothing about the response reveals this number, which is the reason step 5 of the tuning order looks at the force separately (§2-9).

Why the integrator goes negative in problem 3

While the force is on its limit, back-calculation drives the integrator towards the value that makes the demand equal to the limit, $I^* = \tau_{\max} - P - D + (K_i/K_b) e$ (§2-8). Just after the step, P alone is 10 N against a limit of 2.5 N, so I^* is negative, and the integrator follows it down. That is the formula working: the integrator is holding the demand at the limit instead of piling up on top of it.